DSE6111

Session # 1: Introduction to Predictive Modeling

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Course Overview & Objectives

- This course is about developing "forward-looking statistical models." We'll move beyond just describing data to predicting future or unknown observations.
- **Key Learning Objectives:**
  - The ability to manipulate and process available data.
  - The ability to assess the validity and reliability of analytic outcomes.
  - The ability to develop, evaluate, and effectively apply predictive modeling solutions to real-world problems.
  - **ISLRv2 is our core text:** This course structure directly maps to its chapters. This week, we are primarily in Chapter 2, with a forward look into Chapter 3.
- **Software:** We'll use R for exercises and labs are designed using R but, Python is perfectly acceptable for assignments

The Essence of Statistical Learning

What is Statistical Learning?

At its core, statistical learning refers to a vast set of tools for understanding and making sense of complex datasets. It's about building models to predict outcomes or to gain insights into relationships within data. Examples of what tools refer to:

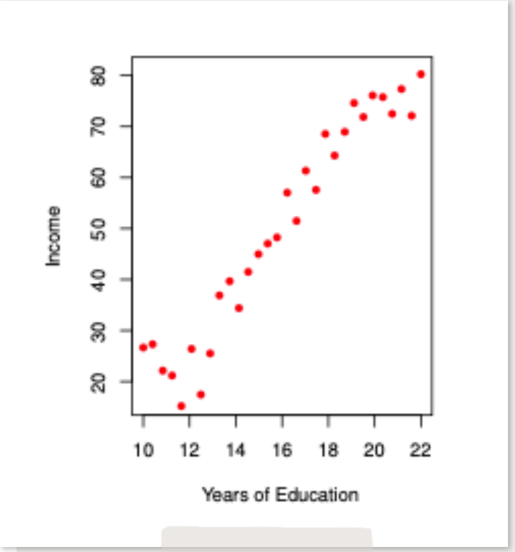
- Linear & Logistic Regression
- Linear Discriminant Analysis
- Decision Trees (Bagging, Random Forests, Bagging)

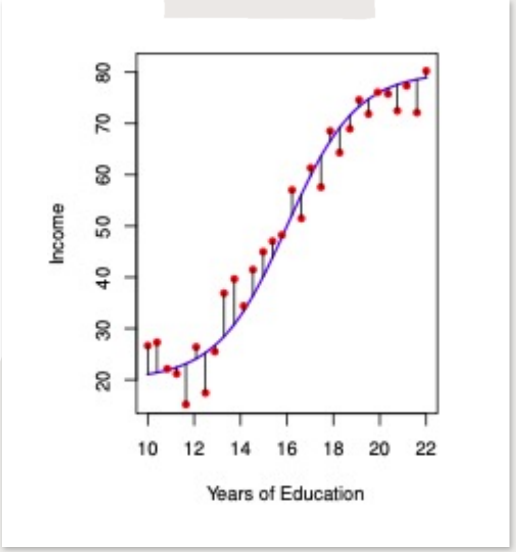
The Fundamental Model: Almost all statistical learning problems can be expressed in a simple, yet powerful, equation:

$$\boldsymbol{Y = f(X) + \varepsilon}$$

- Y is our **response variable** (what we want to predict or understand).
- X represents our **predictor variables** (also known as features, independent variables, or covariates).
- f is some **fixed but unknown function** that relates Y and X. This is the systematic information X provides about Y.
- ε (epsilon) is a **random error term**

In statistical learning, our primary task is to estimate this unknown function f, that connects a set of predictors X, to a response Y. This relationship is almost always effected by some random noise, ε , which we call irreducible error. How we do this, and why, defines much of our course.





Response & Explanatory variables: Roles in Modeling

Explanatory variables (X_1, X_2, X_3): Are manipulated or controlled by the researcher and are termed the cause to explain their effect on the response variable. Also known as Independent Variables, Predictors or Features

Response variable (Y): Is the variable being measured, observed or predicted. Its changes are determined by the influence of explanatory variables. Also known as as Dependent or Target Variable

The response variable is expected to change in response to changes in the explanatory variable. This relationship between explanatory and response variables is crucial in statistical learning as it helps us identify causal links (although correlation is not causation!) and, importantly, predict outcomes

Method	SellerG	Date	Distance	Postcode	Regionname	Propertycount	ParkingArea	Price
SS	Jellis	2016-03-09T00:00:00	2.5		Northern Metropolitan	4019	Carport	
PI	Nelson	2016-03-09T00:00:00	13.5	3042	Western Metropolitan	3464	Detached Garage	840000
S	hockingstuart	2016-03-09T00:00:00	3.3	3206	Southern Metropolitan	3280	Attached Garage	1275000
S	Thomson	2016-03-09T00:00:00	3.3	3206	Southern Metropolitan	3280	Indoor	1455000

Independent Variables

Dependent Variable

Response & Explanatory variables

To determine which variables are explanatory and which are response variables, consider the research objective

1	State	Region	Population	HouseholdIncome	HighSchool	College	Smokers	PhysicalActivity	Obese	NonWhite	HeavyDrinkers	TwoParents	Insured
2	Alabama	S	4.849	43.253	84.9	24.9	21.5	45.4	32.4	30.7	4.3	58.7	78.8
3	Alaska	W	0.737	70.76	92.8	24.7	22.6	55.3	28.4	33.1	8.2	69.6	79.8
4	Arizona	W	6.731	49.774	85.6	25.5	16.3	51.9	26.8	20.8	6.3	62.7	74.7
5	Arkansas	S	2.966	40.768	87.1	22.4	25.9	41.2	34.6	21.7	5	62	71.7
6	California	W	38.803	61.094	84.1	31.4	12.5	56.3	24.1	37.7	6.4	65.3	79.7
7	Colorado	W	5.356	58.433	89.5	37	17.7	60.4	21.3	15.8	6.7	69.9	80
8	Connecticut	NE	3.597	69.461	91	39.8	15.5	50.9	25	22.1	6.3	67	87.7
9	Delaware	NE	0.936	59.878	86.9	31.7	19.6	49.7	31.1	30	6.6	60.4	85.7
10	Florida	S	19.893	46.956	87.1	26.5	16.8	50.2	26.4	23.7	7.2	60.2	70.9
11	Georgia	S	10.097	49.179	85.3	29	18.8	50.8	30.3	39.4	4.7	60.3	72.7
12	Hawaii	W	1.42	67.402	94.8	29.7	13.3	60.2	21.8	75	7.6	70.4	90
13	Idaho	W	1.634	46.767	89.9	23.6	17.2	53.9	29.6	8.1	6.2	74.1	75.9
14	Illinois	MW	12.881	56.797	89.8	38.2	18	52.4	29.4	27.5	6.5	66.2	80.6
15	Indiana	MW	6.597	48.248	87.6	27.5	21.9	44.1	31.8	15.4	5.2	65.4	79.1
16	Iowa	MW	3.107	51.843	92.4	32.2	19.5	47	31.3	8.5	6.5	69.6	87.3

Why Estimate $'f'$? Prediction vs. Inference (1/2)

There are two primary motivations for estimating f, and understanding which one is paramount for a given problem is crucial as it dictates our choice of modeling approach

🎯 Prediction

The primary goal is to accurately predict the output Y for new, unseen input X.
The estimated function f can often be a "black box"— we don't need to know its exact structure, as we primarily care about the accuracy of $\hat{y}$, not necessarily the precise form of $\hat{f}$

Examples:

- Estimating the sale price of property (Regression)
- Predicting if an email is spam (Classification)
- Identifying if a medical image has signs of disease (classification)

🔍 Inference

The primary goal is to understand the relationship between the predictors Y and the response X.
The exact form of $\hat{f}$ is critical because we need to interpret the model.

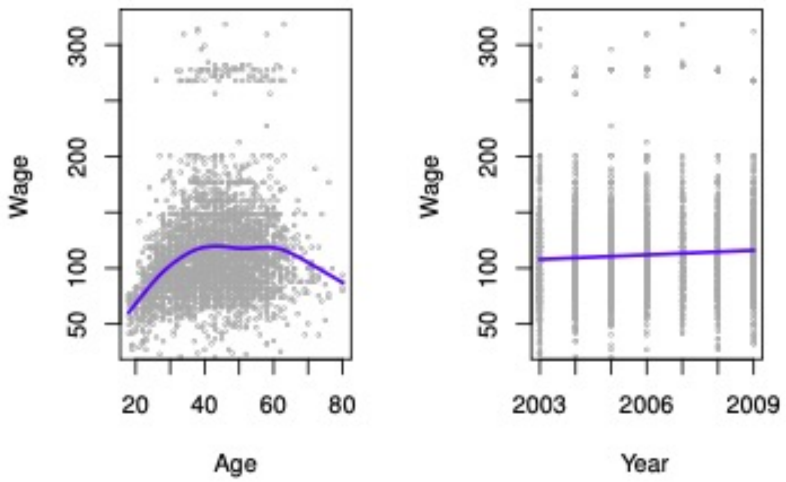
Key Question:

- Which predictors are associated with the response?
- What is the nature (positive/negative, linear/non-linear) of the relationship between Y and each X_j?
- Can the relationship be summarized adequately by a simple equation, or is it more complex?

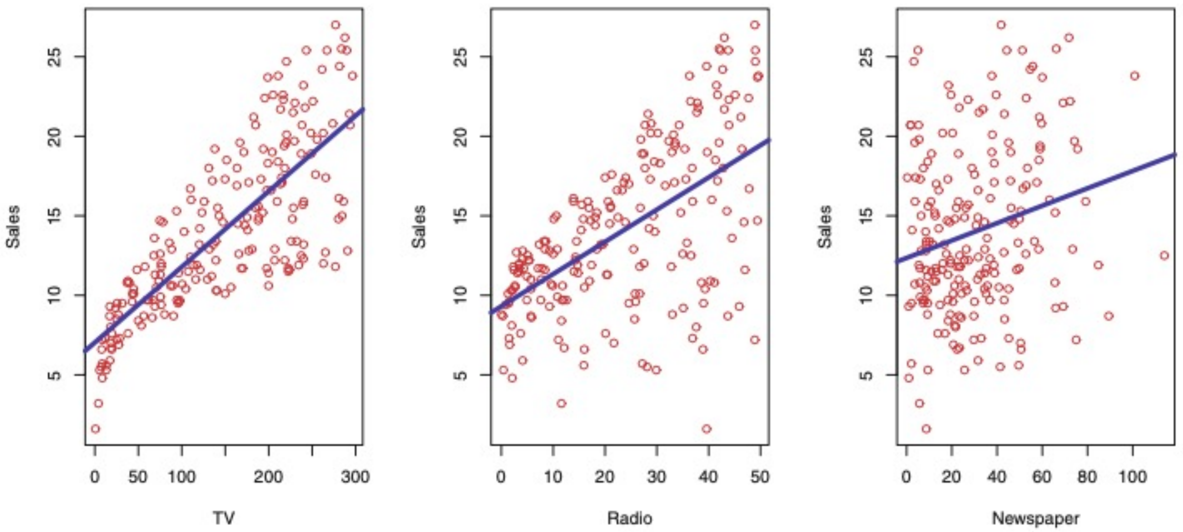
Examples:

- Understanding factors affecting CEO salary.
- Quantifying the impact of advertising budgets on sales

Why Estimate $'f'$? Prediction vs. Inference (2/2)



- **Prediction:** The curved blue line on "wage vs. age" shows a prediction for average wage given age.
- **Inference:** The boxplots of "wage vs. education level" clearly show the *relationship* (inference) between education and wage. We can discuss how a company might want to infer if more education leads to higher wages (inference) versus simply predicting a new hire's wage (prediction)



These simple scatterplots with linear fits instantly convey the idea of understanding relationships between advertising spend and sales.

Statistical Learning vs Machine Learning

These terms are often used interchangeably, and there's significant overlap. However, historically, they emerged from different fields with different emphases

- **Machine Learning (AI Subfield):**
  - **Emphasis:** Large-scale applications and prediction accuracy
  - **Focus:** Algorithms that learn from data to make predictions or decisions without being explicitly programmed for the task. Often less concerned with model interpretability (hence, "black box" models are common)
- **Statistical Learning (Statistics Subfield):**
  - **Emphasis:** Models, their interpretability, precision, and quantifying uncertainty
  - **Focus:** Understanding the underlying data generating process and the relationships between variables
- **The Blurred Lines:** The distinction has become increasingly blurred over time, with a great deal of "cross-fertilization" between the fields. Many techniques are now shared
- **Marketing Appeal:** "Machine Learning" often has stronger marketing appeal, but for this course, we emphasize the rigorous statistical foundations

The Error term: ε and Irreducible Error (1/2)

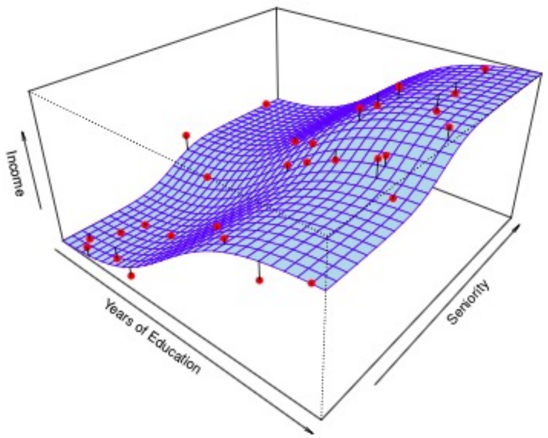
$$Y = f(X) + \varepsilon$$

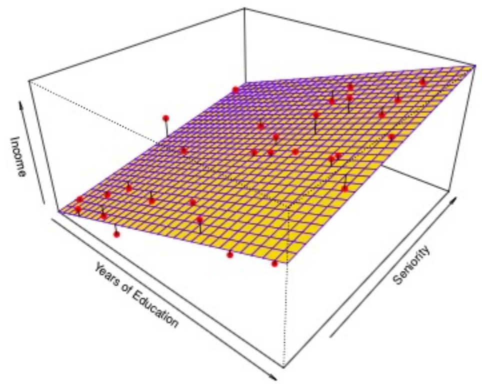
- **The Nature of ε** : ε is the random error term, independent of X and with a mean of zero. It captures everything not accounted for by $f(X)$
- **Components of Prediction Error**: When we make a prediction $\hat{y} = \hat{f}(X)$, the accuracy depends on two quantities:
  - **Reducible Error**: This is the error due to our imperfect estimate of f. We can potentially *reduce* this by choosing a more appropriate statistical learning method or by getting more data
  - **Irreducible Error**: This is the error introduced by ε. No matter how well we estimate f, we cannot predict ε using X. Therefore, this part of the error *cannot be reduced*
- **Mathematical Representation**: The expected squared difference between the predicted and actual value of Y is:

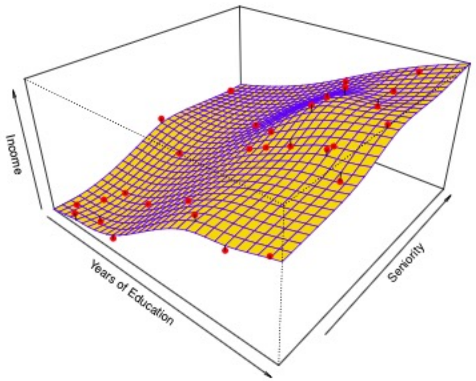
$$E(Y - \hat{Y})^2 = E[f(X) - \hat{f}(X)]^2 + Var(\epsilon)$$

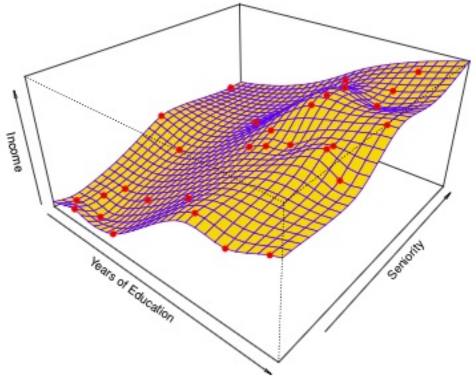
- **Reducible Term** $(E[f(X) - \hat{f}(X)]^2)$: This is what we focus on minimizing through model selection and training
- **Irreducible Term** (Var(ϵ)): This sets an inherent upper bound on the accuracy of our predictions. We can never perform better than this noise level
- **Why is** Var(ϵ) > 0**?** ε can include unmeasured variables that are useful in predicting Y, or inherent unmeasurable variation (e.g., patient's well-being on a given day)

The Error term: ε and Irreducible Error (2/2)









- Red dots are observed data. The **blue surface** is the unknown true function f(X) that genuinely determines income based on education and seniority. Vertical black lines represent the **error term, ϵ.**
- This ϵ is **irreducible error**. It's the inherent noise not explained by X, setting the fundamental limit on prediction accuracy.

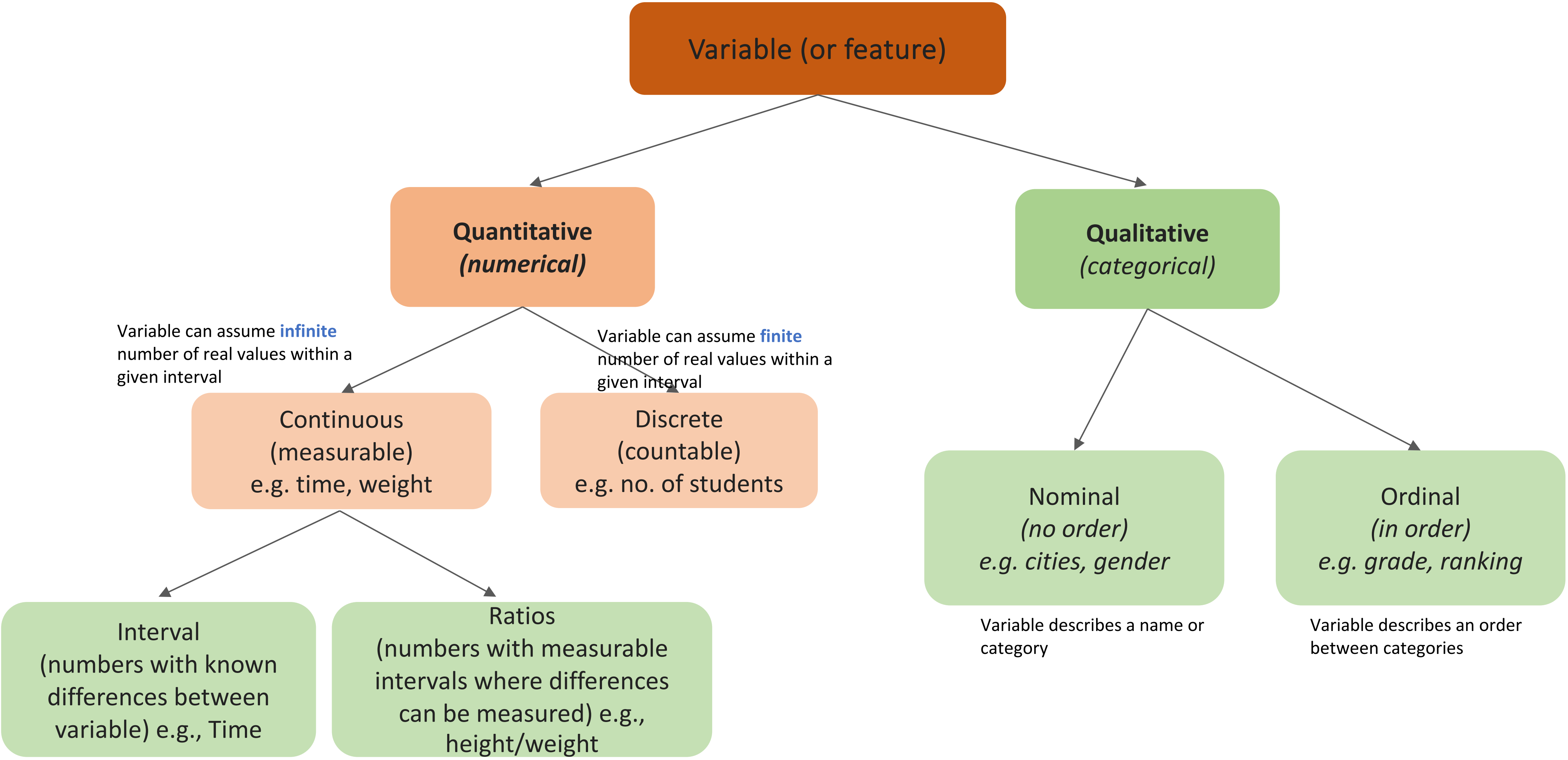
- The **yellow plane** is our estimated linear model, $\hat{f}\,(\boldsymbol{X})$.
- This simple model is too rigid for the true underlying curve, leading to **bias**. Bias is a component of **reducible error** that arises from a mismatch between our model's assumptions and the true function.

- The smooth, curved **yellow plane** is a more flexible model's estimate, $\hat{f}\,(\boldsymbol{X})$.
- This model better approximates the true $\boldsymbol{f(X)}$, significantly **reducing bias** (and thus reducible error) compared to the linear model.

- A rough, wiggly **yellow surface** that perfectly interpolates every red data point).
- This model overfit the training data, learning the noise alongside the signal. This results in high variance as high-test error, as it generalizes poorly to new, unseen data

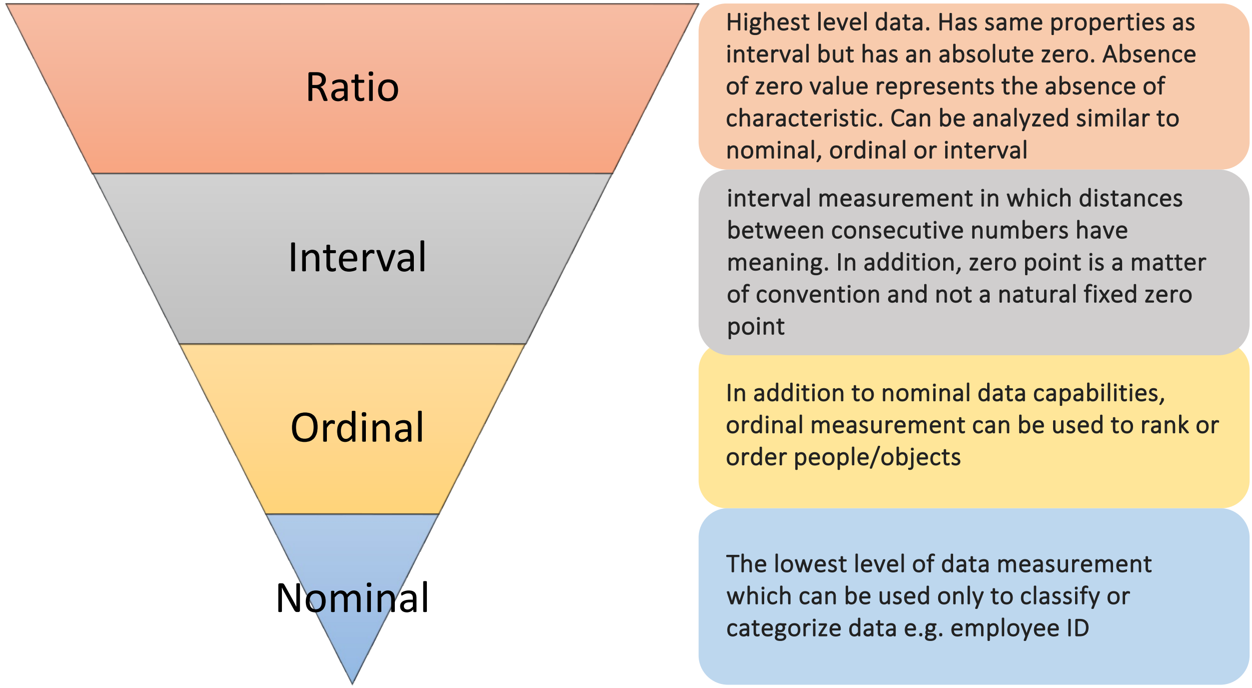
Types of Variables: The building blocks & Measurements

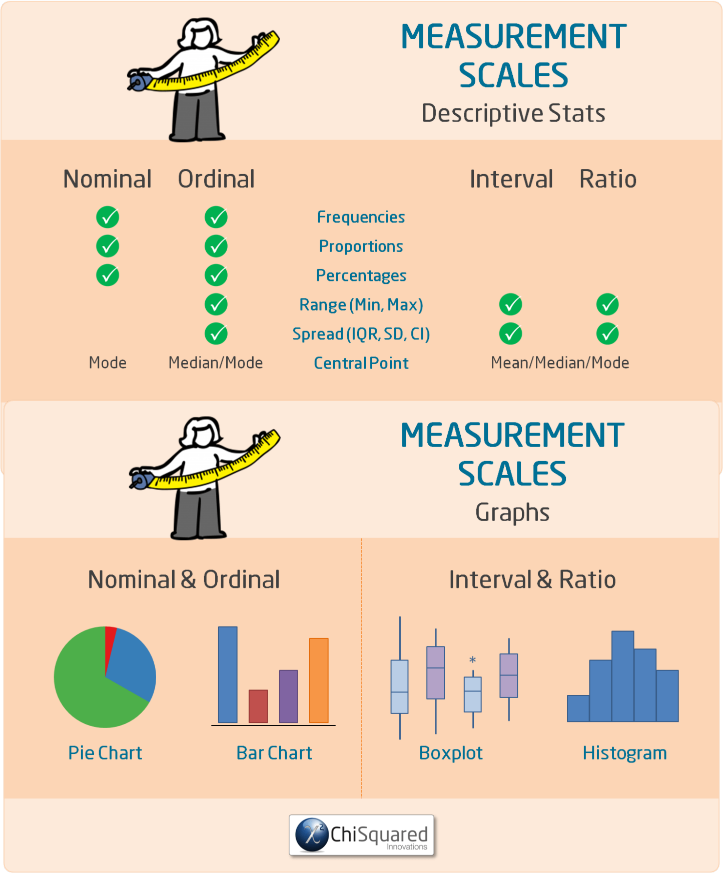
Understanding variable types is fundamental for choosing appropriate models and interpreting results



Data Measurement

The disparate use of numbers can be illustrated by the numbers 40 and 80, which could represent the weights of two objects being shipped, the ratings received on a consumer test by two different products, or football jersey numbers of a fullback and a wide receiver. Although 80 pounds is twice as much as 40 pounds, the wide receiver is probably not twice as big as the fullback! Averaging the two weights seems reasonable, but averaging the football jersey numbers makes no sense. The appropriateness of the data analysis depends on the level of measurement of the data gathered. The phenomenon represented by the numbers determines the level of data measurement.

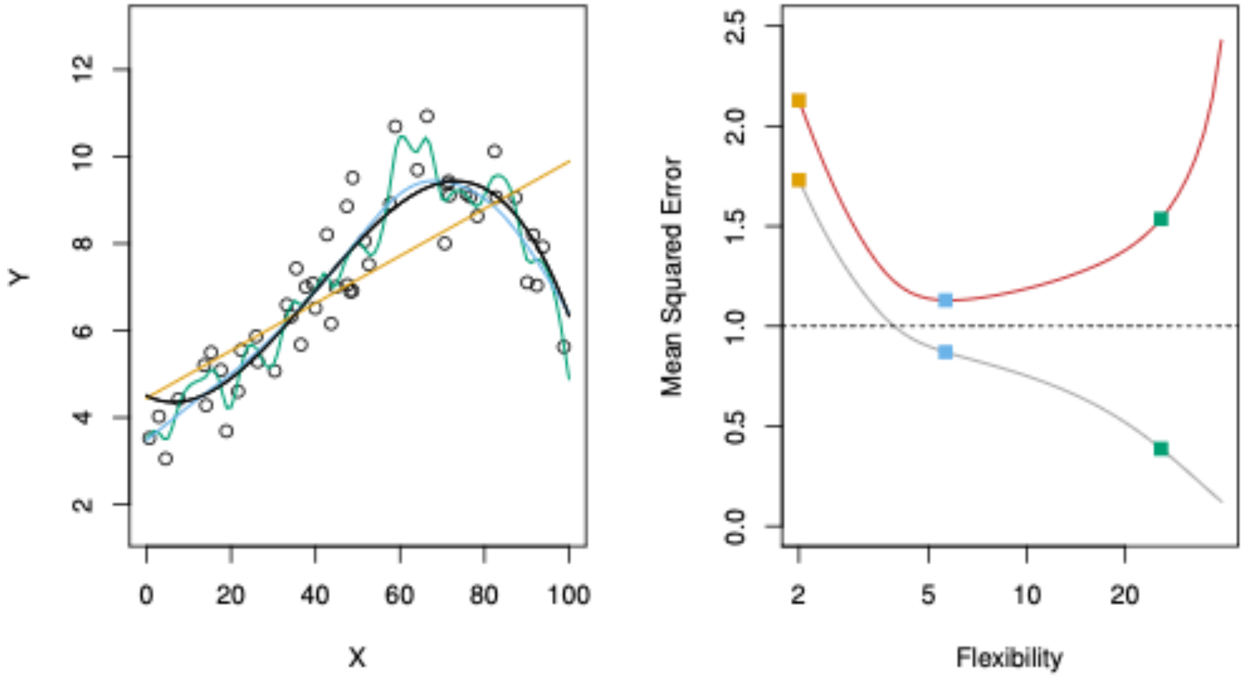




Assessing Model Accuracy: Beyond Training Error

- How do we know if our estimated function $\hat{f}$ is any good? We need to measure its performance.

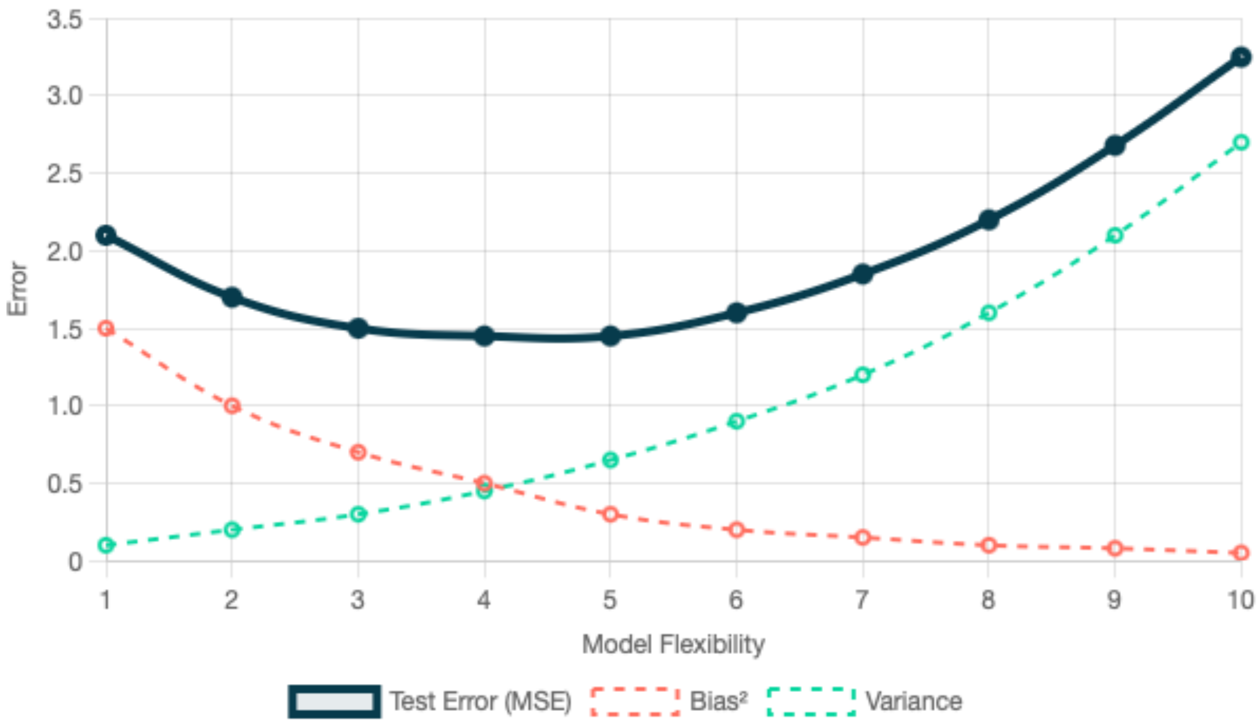
- **The Problem with Training Error**: If we evaluate $\hat{f}$ on the same data used to *train* the model, we get the **training error**. This is almost always optimistically low because the model has seen this data before and specifically tried to fit it well.



- **The Goal: Test Error**: Our real interest lies in how well $\hat{f}$ performs on *new, unseen* observations. This is the **test error**.
- **Why does this matter?**: A model that has low training error but high test-error is said to be **overfitting** the data. It has learned the noise and random fluctuations in the training set, rather than the true underlying signal.
  - Imagine fitting a wiggly line perfectly through noisy data points. It looks great on the training data, but would perform terribly on new data.
- **The U-Shape of Test Error**: As model flexibility increases (e.g., using more complex models or parameters), training error typically decreases monotonically. However, test error often follows a **U-shaped curve**: it decreases initially (as the model captures more signal) and then *increases* (as the model starts overfitting the noise).

Bias-Variance Tradeoff: A Core Principle

- The U-shaped test error curve is a direct consequence of the **bias-variance trade-off**, a fundamental concept in statistical learning.
- **Bias**: The error introduced by approximating a real-world problem (which may be very complex) with a simpler model.
  - Occurs when the model is too simple (e.g., linear model trying to fit a highly non-linear relationship). It consistently misses the true relationship. This leads to **underfitting**
- **Variance**: The amount by which $\hat{f}$ would change if we estimated it using a different training dataset.
  - Occurs when the model is too flexible and sensitive to the specific training data. Small changes in training data lead to large changes in the model. This is characteristic of **overfitting**

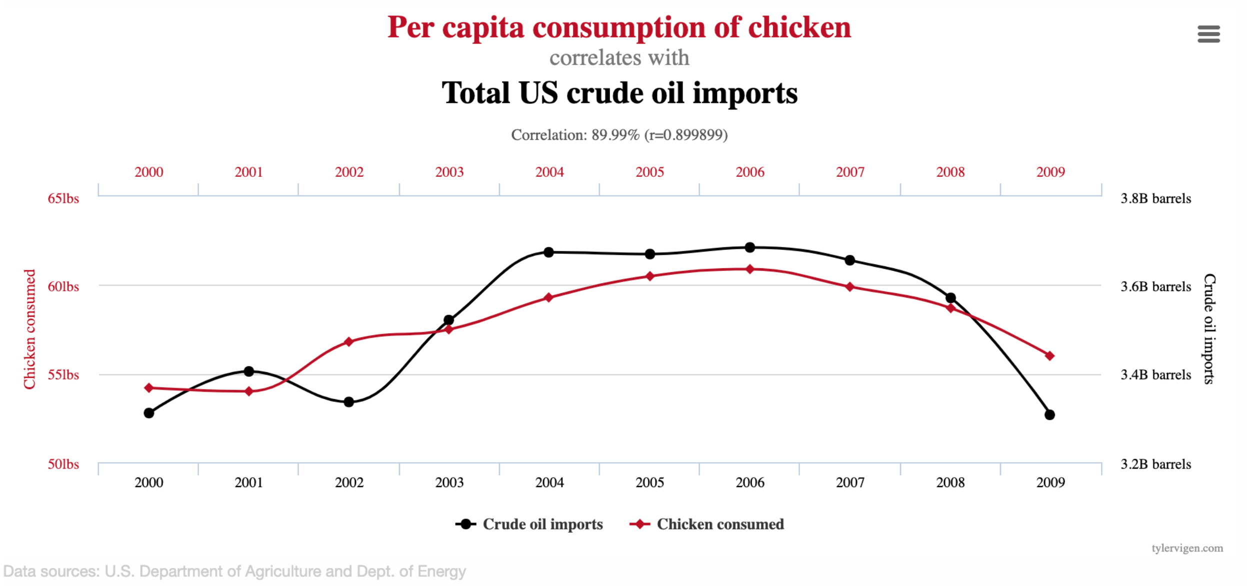


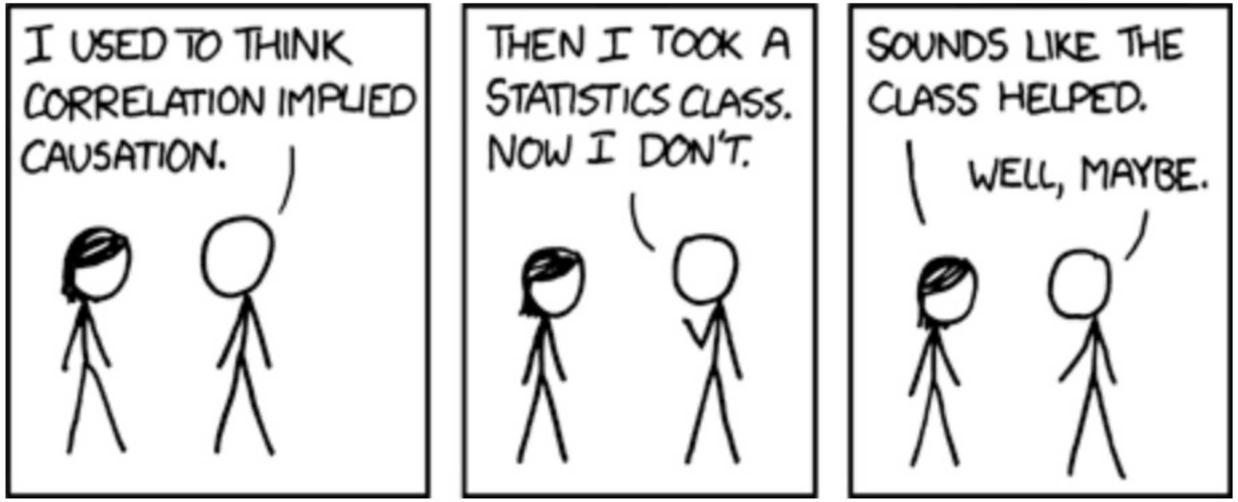
- More flexible models generally have **lower bias** but **higher variance**
- Less flexible models generally have **higher bias** but **lower variance**
- **The Sweet Spot**: The goal is to find the right balance—a model with **low bias AND low variance**—to minimize overall test error. This balance is highly dependent on the specific dataset and problem.

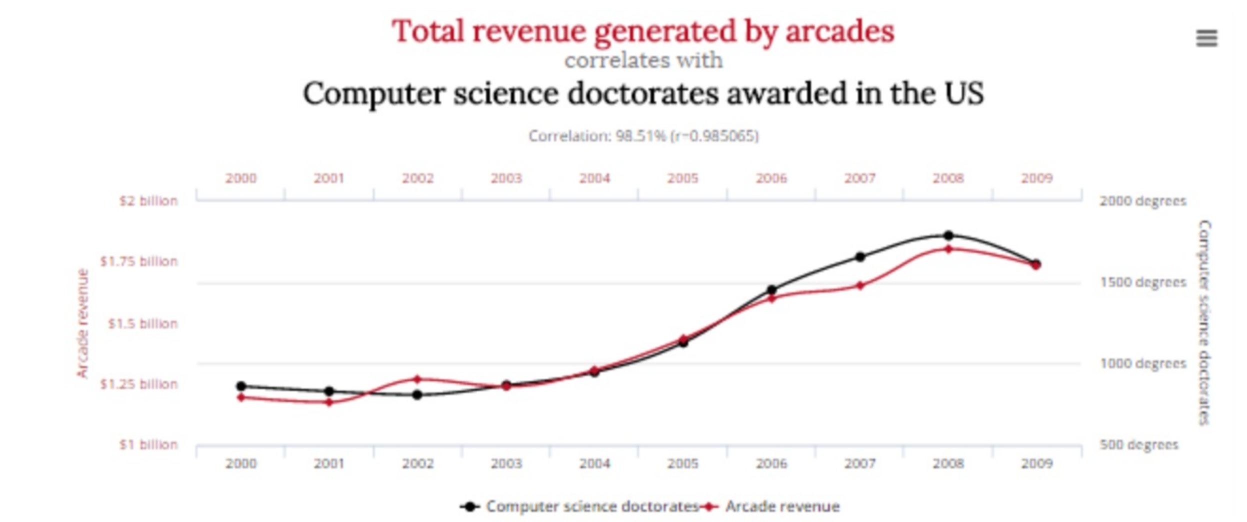
Correlation and Causation

Correlation is not Causation

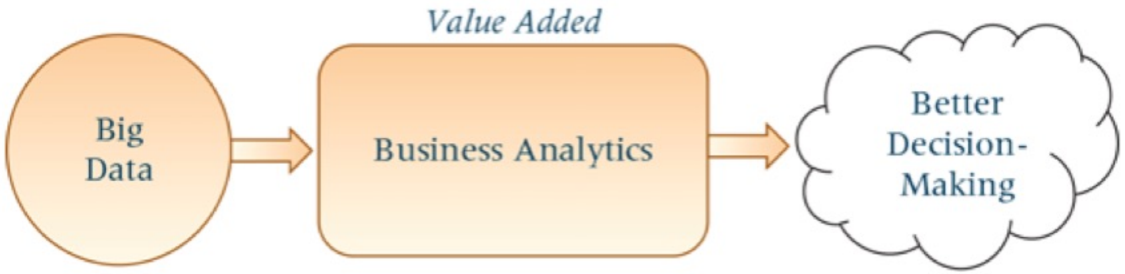
- Correlation is a measure of **strength and direction** of the relationship between variables
- Causation is the **effect or change** of one variable has over another







Prescriptive Statistics v. Predictive Modeling



Predictive Modeling

- It finds relationships in the data that are not readily apparent with descriptive analytics. With predictive analytics, patterns or relationships are extrapolated forward in time and the past is used to make predictions about the unknown.
- Predictive analytics builds and assesses algorithmic models aimed at making empirical rather than theoretical predictions and is designed to predict future or unknown observations.
- Topics in predictive analytics can include regression, time-series, forecasting, simulation, data mining, statistical modeling, machine learning techniques, and others. They can also include classifying techniques, such as decision tree models and neural networks

Prescriptive Modeling

- Predictive analytics builds on the information from predictive analytics to recommend specific sets of actions given a complex set of objectives, requirements, and constraints.
- Prescriptive modeling takes uncertainty into account, recommends ways to mitigate risks, and tries to see what the effect of future decisions will be to adjust the decisions before they are made
- Topics in prescriptive analytics traditionally come from the fields of management science or operations research and are generally those aimed at optimizing the performance of a system such as mathematical programming, simulation, network analysis, and others.